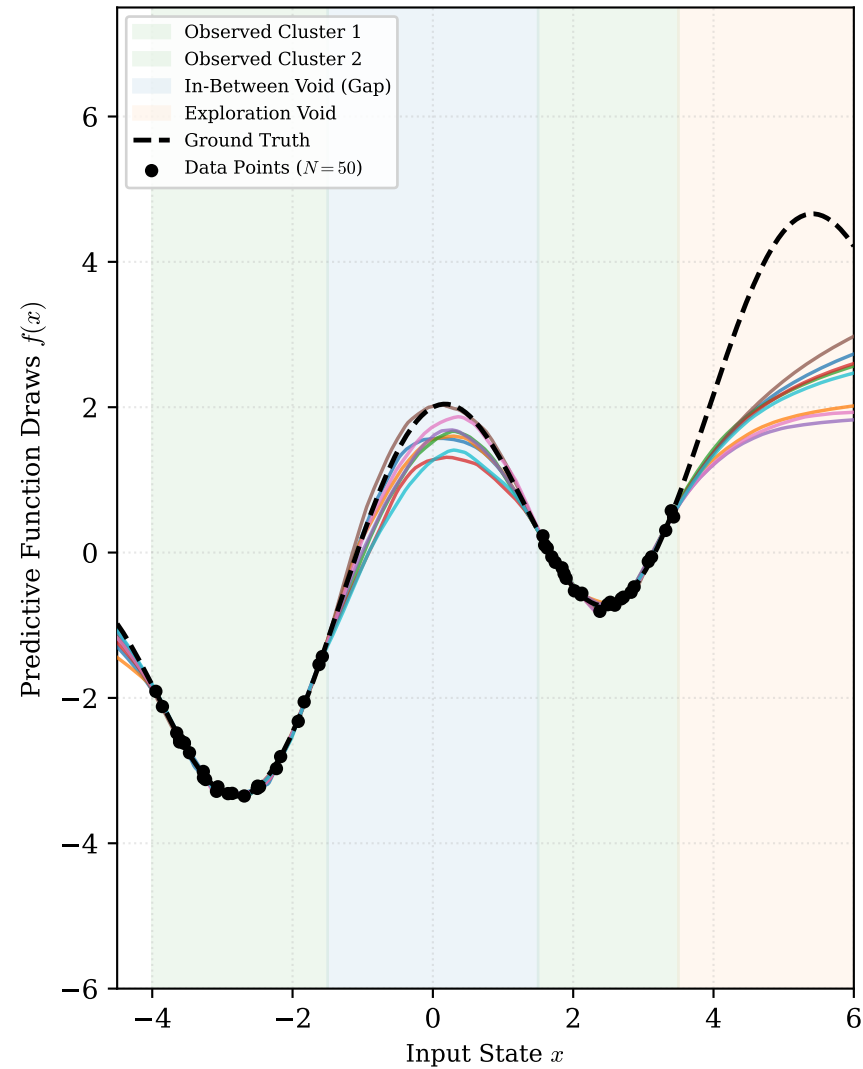
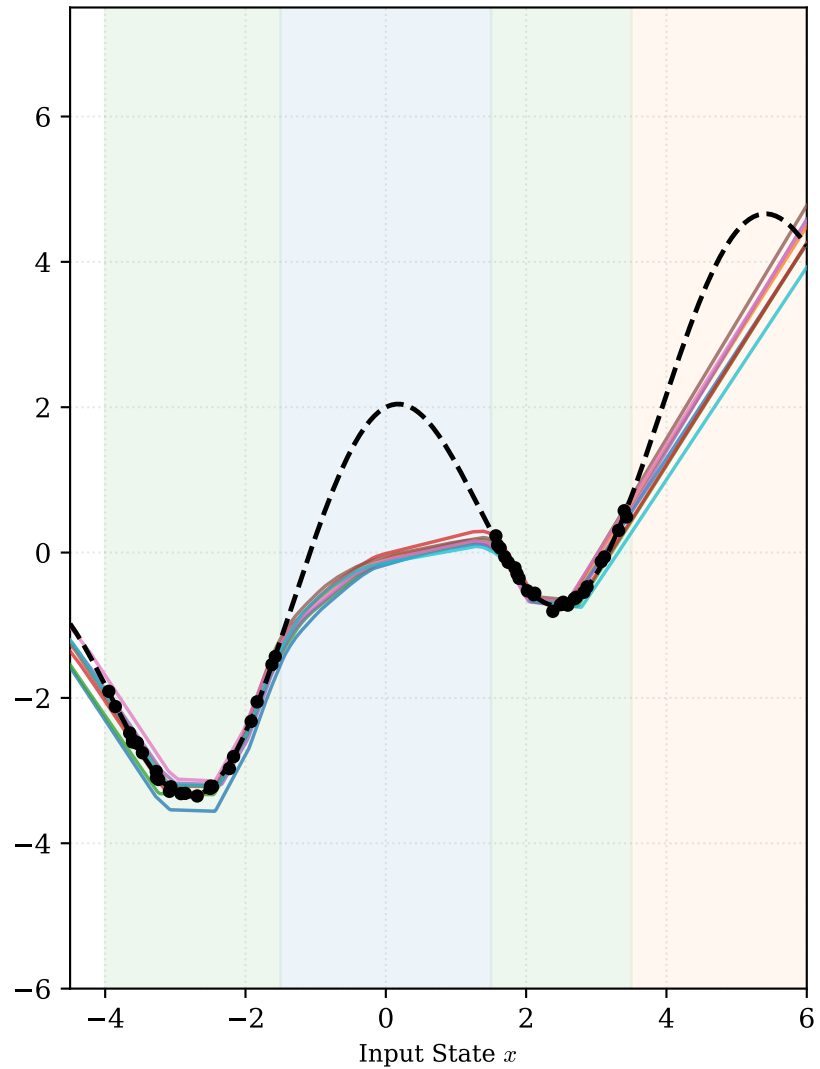


# The Fundamental Mechanism: Why Measure-Space Priors (DP-BNN) Succeed Where Weight-Space Priors Fail [Spaghetti Plots of 8 Individual Posterior Function Realizations Across Dual Voids]

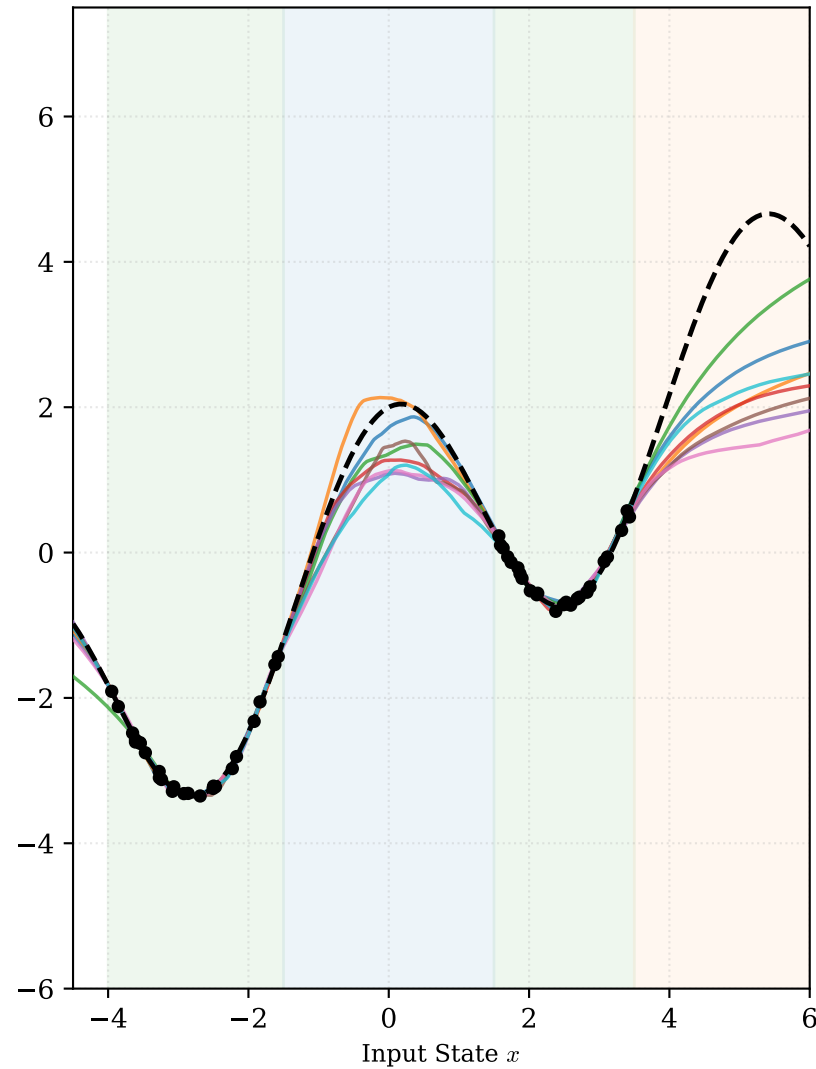
## A. Deep Ensembles



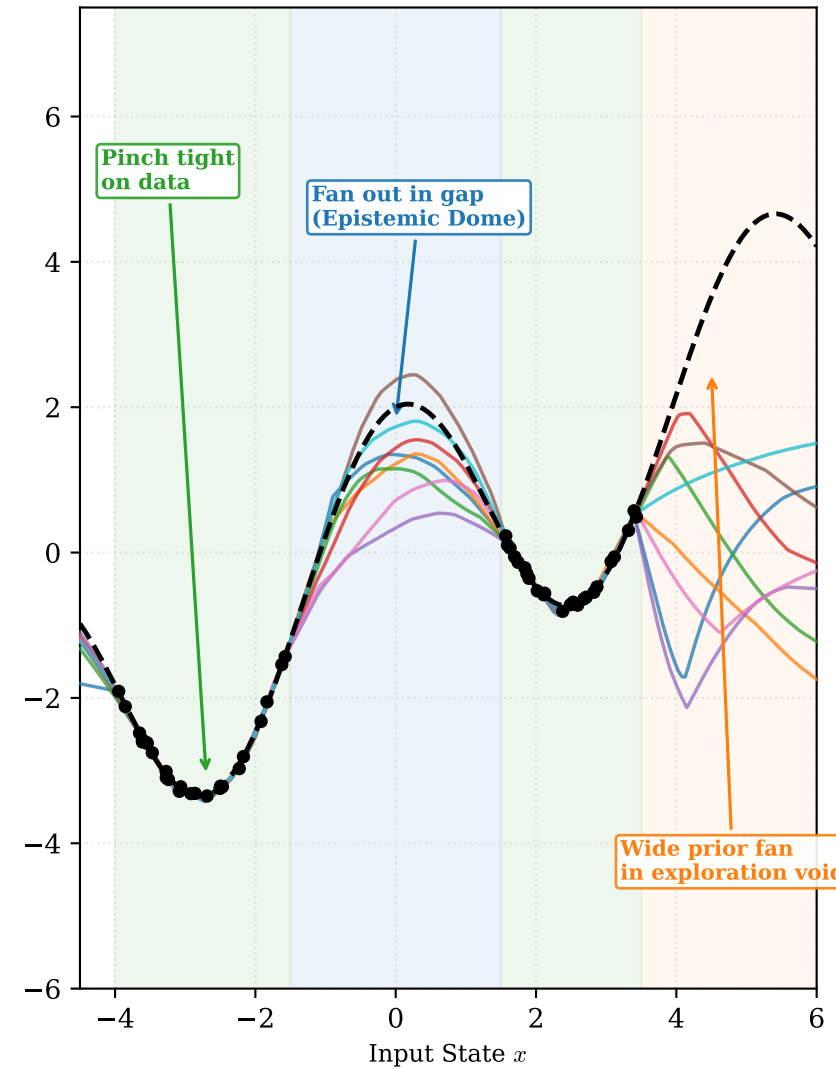
## B. Variational Inference (BBB)



## C. BootDQN + Priors



## D. DP-BNN (Ours)



### Weight-Space (MAP)

$$w \sim \mathcal{N}(0, \sigma^2 I)$$

Failure: Minimal-Norm Collapse  
All members fit identical linear  
interpolants in voids. Zero diversity.

### Weight-Space (MFVI)

$$q(w) = \prod_i \mathcal{N}(\mu_i, \sigma_i^2)$$

Failure: In-Between Collapse  
Weight uncertainty cancels out  
through multi-layer ReLUs.  
Overconfident in gaps.

### Function-Space (Additive)

$$Q = f_\theta(x) + \beta p(x)$$

Failure: Unbounded Drift  
Prior is added to trainable net.  
Unconstrained ReLUs shoot into  
wild negative/unphysical values.

### Measure-Space (Dirichlet)

$$\mathcal{P} \sim \text{DP}(\alpha, P_0)$$

Success: Bayesian Pinch & Fan  
Data gets 96% mass (pinches tight).  
Voids get 100%  $P_0$  (fans out into  
smooth, bounded hypotheses).